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Driver attention based steering effort modulation to prevent accidental lane changes

Abstract

A number of variations may include a vehicle, system and method of modulating power steering or torque from a lane keep assist system based on driver attentiveness to avoid accidental land changes.

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Background/Summary

TECHNICAL FIELD

(1) The field to which the disclosure generally relates to vehicle with electric or hydraulic power steering, or lane keep assist, and systems and methods using the same.

BACKGROUND

(2) Vehicles typically include power steering systems.

SUMMARY OF ILLUSTRATIVE VARIATIONS

(3) A number of variations may include a vehicle, system and method of modulating power steering or torque from a lane keep system based on driver attentiveness to avoid accidental land changes.

(4) Other illustrative variations within the scope of the invention will become apparent from the

detailed description provided hereinafter. It should be understood that the detailed description and specific examples, while disclosing variations of the invention, are intended for purposes of illustration only and are not intended to limit the scope of the invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Select examples of variations within the scope of the invention will become more fully understood from the detailed description and the accompanying drawings, wherein:

(2) FIG. **1** is a flow diagram illustrating a method of modulating power steering assist based on driver attentiveness according to a number of variations;

(3) FIG. **2** is a schematic illustration of a vehicle drive attentiveness monitoring system;

(4) FIG. **3** is a schematic illustration of a vehicle and system for modulating power steering assist based on driver attentiveness according to a number of variations.

DETAILED DESCRIPTION OF ILLUSTRATIVE VARIATIONS

(5) The following description of the variations is merely illustrative in nature and is in no way intended to limit the scope of the invention, its application, or uses.

(6) In level 2-3 ADAS systems or even in case of full manual driving, driver monitoring input can be used to reduce assist in steering system when the driver attention is not on the road ahead. This will reduce chances of driver inadvertently leaving the current lane due to hands being on the wheel causing vehicle yaw movement due to full availability of power assist. If the assist in this situation is reduced, amount of torque required to cause yaw movement will be substantially more.

(7) When a driver (in manual mode driving) is distracted and looses focus on the road, the ability to keep steering wheel on center (when the road is straight) is also lost. The weight of the hand/s can cause the steering system assist to kick in and cause yaw movement of the vehicle leading it to veer into another lane. A vehicle as describer hereafter may include a lane keep assist (LKA) system including an algorithm that uses the inputs of measured steering wheel torque and position of the vehicle relative to lane boundaries to determine if lane keeping steering torque should be produced. If the inattentive driver is applying sufficient steering wheel torque input then there could be instances where the lane keeping assist torque is not sufficient to avoid the vehicle crossing over the lane boundary. LKA systems most often use a forward-looking camera to identify lane boundary markings/painted lines. Very often road and/or environmental conditions (e.g., snow, heavy rain) make it difficult to identify the lane boundaries so the lane keeping system has a temporary loss of confidence in the lane boundary and is unable to generate the lane keeping steering torque request if needed.

(8) By accepting an input from driver monitoring system, which can tell us levels of attention that the driver has or just provide a flag confirming that the driver is being assessed as “inattentive”, the steering system will reduce assist levels. With reduced assist levels, torque caused by the weight of a hand or higher torque due to tensed uncontrolled muscles will not generate enough assist from the power assist mechanism to generate sufficient yaw movement in the vehicle. A second implementation could be an enhancement to existing lane keeping assistance equipped vehicles. The input from the driver monitoring system would be used to supplement the lane position of the vehicle to provide an “early lane keeping steering torque of smaller magnitude” to counteract the inattentive driver from inducing the unwanted yaw or lateral response.

(9) The vehicle may be installed with a form of driver monitoring system which can distinguish between an attentive driver and an inattentive driver. This may be in the form of cameras looking at the driver's face/body or hands based attention systems. The output of this system can be either a simple flag confirming whether the driver is attentive or not attentive along with the time stamp at which the event occurred. In addition the system may also output a varying scale of attentiveness.

This information will be sent out over available communication system to which the steering system is connected. The steering system ECU may monitor this signal and when appropriate (when the driver is deemed inattentive) will switch the power assist available for the given amount of torque being input from the steering column to be lower than normal or turn it off completely. This may be done in a controlled manner so as to prevent kick in the system and will also be able to bring the assist back gradually (which will be tunable) when the driver regains attention. In a similar method, a steering system that is supporting lane keeping assistance could modify the lane keeping steering torque overlay based upon the received driver attentiveness input.
(10) This can be standard feature in all vehicles that have some form of driver monitoring and also have an electric power assisted steering system or hydraulic PAS with some ECU based control for the assist boost.
(11) A number of variations may include the integration of driver monitoring and steering system effort assist with reduction of assist based on driver attention, and resumption of assist based on driver regaining attention.
(12) Referring now to FIG. **1**, a number of variations may include a method and/or system for carrying out the method including increasing the force a driver of a vehicle must exert on a steering wheel or steering interface in response to a determination that the driver is inattentive to the task of driving so that unwanted yaw or vehicle movement is avoided or reduced. The method may include activating full power steering assist and/or lane keep assist **100**. For example, the activation of full power steering assist and/or lane keep assist may occur upon starting the vehicle or by pressing button or other means activating the same. Next, collecting data pertaining to driver attentiveness **102**, and analyze the data for signs of driver inattentiveness **104**. Then determining if the driver has not become inattentive to the task of driving **106**, and if so repeating steps **102**, **104**, and **106**. If not, then reducing or stopping power steering, or applying lane keep assist torque to counteract unwanted yaw or lateral movement of the vehicle **108**. Thereafter, collecting data pertaining to driver attentiveness **110**. Thereafter, analyzing data for signs of driver attentiveness **112**. Determining if the driver has become attentive to the task of driving **113**. If the driver is not attentive, then repeating steps **110** and **112**. If the driver is attentive, then applying full power steering or reducing lane keep assist torque **114**. Full power steering may be restored in an incremental fashion. Thereafter, the method repeats from step **102**.
(13) FIG. **2** illustrates the view of a driver through a front windshield **203** of a vehicle traveling on a driving surface **202** which may include one or more lane markers **210**. The vehicle is on a path indicated by arrow **206** so that the vehicle stays within lane markers **210**. Even though one or more hands **216** of the driver may be on the vehicle steering wheel **212** the driver may become inattentive, for example, by looking off to the side of the road or reaches in the cabin for an object and causes the steering wheel **210** to move or rotate in the direction indicated by arrow **214**. The movement or rotation of the wheel **210** may cause the vehicle to stray to the path indicated by arrow **208** and cross over the lane maker **210**. To prevent unwanted yaw or lateral movement the vehicle may have a camera **218** which may be positioned anywhere in the vehicle such as the passenger cabin. In one variation, the camera may be placed on the steering wheel **212**. In another variation an alternative or additional camera **220** may be positioned on a rear-view mirror **204** of the vehicle. The camera **218**, **220** may be utilize to collect data pertaining to the driver attentiveness. A controller as described hereafter may be used to carry out the methods described herein to prevent unwanted yaw or lateral movement the vehicle.
(14) Referring now to FIG. **3**, an illustrative variation of a vehicle equipped with hardware that allows it to carry out at least some of the methods disclosed herein is shown. A vehicle **301** is equipped with wheels **302** and a handwheel **303** for turning the wheels **302** via a pinion **304** that engages a rack **305** that is constructed and arranged to turn the wheels **302**. In the illustrative variation shown, the handwheel **303** is equipped with a hand wheel torque sensor **306** and a hand wheel angle sensor **307** so that any turning of the handwheel may produce sensor data that may be

communicated to or accessed by a controller **308**. Although, in this illustrative variation, the controller **308** is shown onboard the vehicle, the controller may also be located somewhere apart from the vehicle and communicated with wirelessly by the sensors or the vehicle. The pinion **304** may be equipped with a pinion torque sensor **309** so that any turning of the pinion may be observed by or communicated to the controller **308** and utilized by the methods described herein. In the illustrative variation shown, the rack **305** is equipped with a rack force sensor **310** so that any rack forces detected during driving may be observed by or communicated to the controller **308** and utilized by the methods described herein. The vehicle may have a steering shaft **314** connecting the steering wheel or steering interface **303** to the pinion **304**. An electric power steering assist or hydraulic power steering device **316** may be connected to the shaft **314** to assist the drive in steering the wheels of the vehicle by reduce the force or torque the driver would need to apply to the steering wheel or steering interface **303** if the power steering device **316** was not present. Also shown in this illustrative variation, the wheels **302** may be equipped with road wheel sensors so that any road wheel data detected during driving may be observed by or communicated to the controller **308** and utilized by the methods described herein. Additionally, in the illustrative variation shown, cameras **312** are located near the wheels **302**, though at least one of the cameras **312** may be located elsewhere in other illustrative variations. The cameras **312** may be used in conjunction with any sensor on the vehicle **301** that aids in monitoring vehicle travel or usage data at least for the purposes of the methods described herein. Another controller **318** may be provided and may include a processor **320**, memory **322**, wherein the instructions stored in the memory **322** are executable by the processor to modulate the power steering from providing full power steering to less than full power steering or to completely stop the power steering device from assisting the driver to steer the vehicle; or to apply lane keep assist torque to the steering wheel or steering interface **303** to counteract unwanted yaw or lateral movement of the vehicle that could otherwise by created by an inattentive driver. A separate controller similarly configured as controller **318** may be used to control the lane keep assist function and to apply lane keep assist torque to the steering wheel or steering interface **303** counteract unwanted yaw or lateral movement of the vehicle that could otherwise occur by an inattentive driver.

(15) The above description of select variations within the scope of the invention is merely illustrative in nature and, thus, variations or variants thereof are not to be regarded as a departure from the spirit and scope of the invention.

Claims

1. A method comprising determining if an unconscious torque application by an inattentive driver of a vehicle is turning the vehicle in the direction opposite of a road curvature ahead of the vehicle, and in response to a determination that an unconscious torque application by the inattentive driver is turning the vehicle in the direction opposite of the road curvature ahead of the vehicle completely stopping an electric or hydraulic power steering of the vehicle to counteract unwanted yaw or vehicle movement.

2. A method comprising: (a) activating full power steering and/or lane keep assist in a vehicle; (b) collecting data pertaining to driver attentiveness; (c) analyzing the data for signs of driver inattentiveness; (d) determining if the driver has not become inattentive to the task of driving, and if so repeating acts (a), (b), and (c), and if not, then completely stopping power steering to counteract unwanted yaw or lateral movement of the vehicle if an unconscious torque application by the inattentive driver of the vehicle is turning the vehicle in the direction opposite of a road curvature ahead of the vehicle; (e) thereafter, collecting data pertaining to driver attentiveness; (f) analyzing data for signs of driver attentiveness; (g) determining if the driver has become attentive to the task of driving, and if not, repeating acts (e), (f) and (g), and if the driver is attentive, then applying full power steering; (h) thereafter, repeating acts (a)-(g).

3. The method as set forth in claim 2 wherein in step (g) applying full power steering is accomplished in an incremental fashion.
4. The method as set forth in claim 2 wherein the power steering is electric power steering assist.
5. The method as set forth in claim 2 wherein the power steering is hydraulic power steering.
6. The method as set forth in claim 2 wherein act (g) comprises applying lane keep assist torque to counteract unwanted yaw or lateral movement of the vehicle.
7. A system for using in a vehicle comprising a power steering system or a lane keep assist system, a controller comprising at least one processor, memory and instructions stored in the memory executable by the processor causing the system to carry out the acts comprising: (a) activating full power steering and/or lane keep assist in a vehicle; (b) collecting data pertaining to driver attentiveness; (c) analyzing the data for signs of driver inattentiveness; (d) determining if the driver has not become inattentive to the task of driving, and if so repeating acts (a), (b), and (c), and if not, then completely stopping power steering to counteract unwanted yaw or lateral movement of the vehicle if an unconscious torque application by the inattentive driver of the vehicle is turning the vehicle in the direction opposite of a road curvature ahead of the vehicle; (e) thereafter, collecting data pertaining to driver attentiveness; (f) analyzing data for signs of driver attentiveness; (g) determining if the driver has become attentive to the task of driving, and if not, repeating acts (e), (f) and (g), and if the driver is attentive, then applying full power steering; (h) thereafter, repeating acts (a)-(g).
8. The system as set forth in claim 7 wherein in act (g) applying full power steering is accomplished in an incremental fashion.
9. The system as set forth in claim 7 wherein act (g) comprises reducing or stopping power steering to counteract unwanted yaw or lateral movement of the vehicle.
10. The system as set forth in claim 9 wherein the power steering is electric power steering assist.
11. The system as set forth in claim 9 wherein the power steering is hydraulic power steering.
12. The system as set forth in claim 7 comprising applying lane keep assist torque to counteract unwanted yaw or lateral movement of the vehicle.
13. A system for using in a vehicle comprising a power steering system or a lane keep assist system, a controller comprising at least one processor, memory and instructions stored in the memory executable by the processor causing the system to carry out the acts comprising: determining if a driver of the vehicle is inattentive and if so, monitoring the road conditions ahead of the vehicle; determining if an unconscious torque application by the inattentive driver is turning the vehicle in the direction of a road curvature ahead of the vehicle, and if so, then not reducing or modulating the steering assist levels; and determining if an unconscious torque application by the inattentive driver is turning the vehicle in the direction opposite of the road curvature ahead of the vehicle, and if so, then completely stopping steering assist.